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Chitosan-based double network hydrogel loading herbal small molecule for accelerating wound healing

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ABSTRACT

Skin injuries are one of the most common clinical traumas worldwide, and wound dressings are considered to be one of key factors in wound healing. Natural polymer-based hydrogels have been developed as ideal materials for a new generation of dressings due to their excellent biocompatibility and wetting ability. However, the inadequate mechanical performances and lack of efficacy in promoting wound healing have limited the application of natural polymer-based hydrogels as wound dressings. In this work, a double network hydrogel based on natural chitosan molecules was constructed to enhance the mechanical properties, and emodin, a herbal natural product, was loaded into the hydrogel to improve the healing effect of the dressing. The structure of the chitosan–emodin network formed by Schiff base reaction and microcrystalline network of biocompatible polyvinyl alcohol endowed hydrogels with excellent mechanical properties and ensured its integrity as wound dressings. Moreover, the hydrogel showed excellent wound healing properties due to the loading of emodin. The hydrogel dressing could promote cell proliferation, cell migration, and secretion of growth factors. The animal experimental results also demonstrated that the hydrogel dressing facilitated the regeneration of blood vessels and collagen and accelerated wound healing.

1. Introduction

Skin injury is a common clinical disease of sudden trauma, and wound management is a huge challenge worldwide [1,2]. Wound healing disorder and its related complications bring great pain and economic burden to patients. In the course of clinical treatment, wound dressing could temporarily replace the injured skin and protect the wound, which is considered to be one of the keys to wound healing [3–5]. Traditional wound dressings, such as gauze and cotton pads, are not as effective as expected in promoting wound healing because they could not provide a favorable healing environment. Wound dressing should not only cover the wound but also facilitate the healing process of the wound [6]. Ideally, wound dressing should have the following characteristics: (1) biocompatibility to ensure that the dressing does not cause harm to body; (2) moisturizing ability to maintain the moist environment of the wound and the absorption ability of the wound exudate; (3) good mechanical properties to ensure the integrity of its structure and not be invaded by external infection; and (4) ability to stimulate the growth of blood vessels and connective tissue to accelerate wound healing [7–10].

Hydrogel is a kind of three-dimensional material constructed by the hydrophilic polymer network capable of holding a large amount of water. Due to excellent biocompatibility, permeability, and tissue-like property, hydrogels have been extensively studied and applied in biomedicine [5,7,11]. Compared with traditional gauze or new natural biological dressing in clinical application, such as xenograft skin, hydrogel could absorb a certain amount of wound exudate and provide a moist healing environment for the wound. Therefore, hydrogel is considered to be the most competitive modern dressing [12–17]. However, traditional hydrogels tend to be brittle and are prone to material breakage under pressure, leading to external bacterial infection. For improvement of the mechanical properties of hydrogels, Gong et al. proposed the preparation strategy of double network hydrogels, which greatly improved the toughness of hydrogels [18]. On this basis, Suo et al. introduced Ca^{2+} and alginate network into the covalent cross-linked polyacrylamide network and realized the self-recovery property

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of hydrogel, and the fracture toughness reached 9000 J/m^2 [19]. Therefore, great progress has been made in preparing new hydrogel dressings by using double network hydrogels. Most of the double network hydrogels are formed by polymerization of small molecular monomers, and acrylamide monomers have been widely used [20]. However, some unreacted monomers inevitably remain during the process of preparing hydrogels, and the residual acrylamide monomer has toxicity, which poses potential safety risks for the application of wound dressing. The promising strategy to obtain good biosecurity is to choose biological macromolecules for the synthesis of hydrogels [21,22]. However, constructing double network hydrogels with sufficient mechanical properties based on macromolecules with excellent biocompatibility remains a challenge.

Chitosan (CS) is a kind of natural cationic polysaccharide extracted from crustaceans. It has good bacteriostatic, biosafety, and biodegradability, and it has attracted considerable attention in the field of biomedicine [23]. Therefore, CS molecules were selected as one of the components of the construction of double network hydrogels [24–28]. On the other hand, traditional Chinese medicine and its associated derivatives, showing anti-inflammatory, antibacterial, stimulating cell activity, and other effects, jointly promote wound healing and have become a common drug in the clinical treatment of skin injury [29,30]. Emodin, a herbal small molecule, could be found in Chinese medicines, such as rhubarb, *Polygonum multiflorum*, and *Polygonum cuspidatum*; it has a wide range of pharmacological effects, such as anticancer, anti-inflammatory, antiviral, and antibacterial effects [31–39]. The extracts of this traditional Chinese medicine have been proven to have certain associations with the wound healing process. For example, Song et al. demonstrated that emodin could induce type I collagen synthesis in Hs27 human dermal fibroblasts [40]. Yang et al. confirmed the antibacterial and anti-inflammatory potential of Angelica Dahurica and rhubarb to promote the healing of *Staphylococcus aureus*-infected wounds [41]. Radha et al. showed that emodin could upregulate urokinase-type plasminogen activator and plasminogen activator inhibitor-1 and promote human fibroblast growth [42]. Moreover, the amino group of CS and two carbonyl groups of emodin could react with Schiff base, forming a dynamic covalent crosslinking effect, thus constructing the network of hydrogels.

Polyvinyl alcohol (PVA) is another major component used in double network hydrogels. It is a hydrophilic, nontoxic, biocompatible, and biodegradable substance that has been commonly used in biomedical fields, such as drug delivery, tissue engineering, and wound healing. PVA could be crystallized and crosslinked by stacking polymer chains. In this study, highly crystalline polymer films were first formed by dewatering a mixed aqueous solution of PVA and CS. Then, the polymer film was immersed in an aqueous solution containing emodin. On the one hand, the addition of water molecules could destroy the crystalline structure of the polymer film, thus reducing the crystallinity of PVA and forming a loose microcrystalline network. On the other hand, emodin reacted with CS as the Schiff base to form a second network on the basis of the PVA microcrystalline network. The results of scanning electron microscopy (SEM) and X-ray diffraction (XRD) confirmed that the PVA crystallization and the dynamic covalent crosslinking of CS–emodin were successfully constructed inside the hydrogel. The mechanical properties demonstrated that with the increase in emodin concentration, the tensile strength of the hydrogel increased from 325 kPa to 1070 kPa, and the toughness increased by six times, which also indicated that the second network structure could greatly improve the mechanical properties of the hydrogel. Although the injectable hydrogels showed excellent therapeutic effects on wounds, providing an effective approach for clinical wound management [43–45], our gels exhibited higher mechanical strength, elastic modulus and toughness, which was less prone to damage during treatment process. Moreover, it was also easier to maintain the integrity of the structure in the process of dressing change, which would simplify the process of dressing change. As for Emodin, it was not only a component of the second network but also an effective component in promoting wound healing. The results of drug release experiment showed that the abundant emodin molecules in the hydrogel network could achieve rapid release, providing a basis for subsequent wound healing. At the cellular level, the hydrogel dressing could promote cell proliferation, cell migration, and secretion of growth factors. Animal experiments were performed, and the results demonstrated that the hydrogel dressing effectively accelerated wound healing by facilitating the regeneration of blood vessels and collagen.

2. Experimental section

2.1. Materials

CS and PVA were purchased from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). Emodin was provided by Yuanye Bio-Technology Co., Ltd. (Shanghai, China). Na_2CO_3 was obtained from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). Cell culturing medium (Dulbecco's modified Eagle medium, DMEM) was bought from Biological Industries (Israel), and FBS was purchased from Gibco. Normal human dermal fibroblasts (NHDFs) were purchased from BeNa Culture Collection (Beijing, China). Cell Counting Kit-8 (CCK8) assay was provided by APExBIO (Huston, USA). The cell viability of NHDFs in hydrogels was evaluated by Calcein/PI Live/Dead Viability/Cytotoxicity Assay Kit, which was obtained from Beyotime (Shanghai, China). MonScript RTIII Mix and MonAmp ChemoHS qPCR Mix were bought from Monad Biotech Co., Ltd. (Suzhou, China). Hematoxylin–eosin (HE) staining kit was provided by BaSo (Zhuhai, China). Masson's trichrome staining kit was purchased from Solarbio Science & Technology Co., Ltd. (Beijing, China). CD31 antibody was obtained from Abcam (Cambridge, USA). Ki67 and α-smooth muscle actin (α-SMA) antibodies were bought from Affinity (Ohio, USA). Sprague–Dawley (SD) rats aged 5–6 weeks were purchased from Changsheng Biotechnology Co., Ltd. (Liaoning, China). All animal experiments were approved by the Animal Experiment Ethics Committee of Anhui Medical University (no. LLSC20220731).

2.2. Preparation of PVA/CS/emodin hydrogel

Double network hydrogels containing emodin were prepared by a facile two-step process. In the first step, 1 g of PVA and 0.2 g of CS were added into 10 g deionized water and heated at 95 °C under stirring for 2 h to form a uniform and viscous transparent solution. The small bubbles dissolved in the solution were removed by ultrasound, and then the solution was heated at 60 °C to remove water slowly to form a transparent polymer film. In the second step, the obtained polymer film was immersed in emodin solutions of different concentrations. Considering emodin is insoluble in water, 0.1 M Na_2CO_3 was first prepared to promote the dissolution of emodin. Then, the concentrations of emodin in the solution were set at 0, 0.1, 1, 10, and 100 mg/L separately. After the polymer film was soaked in emodin solution for 12 h, PVA/CS/emodin double network hydrogels were obtained. The hydrogels were divided into five groups in accordance with the concentration of emodin solution: PVA/CS/E_0, PVA/CS/$E_{0.1}$, PVA/CS/E_1, PVA/CS/E_2, and PVA/CS/E_3 groups.

2.3. Characterization

SEM (Phenom Pro X, Netherlands) was used to observe the cross-section morphology of the double network hydrogels. For sample preparation, the hydrogel was first rapidly frozen with liquid nitrogen to fix its shape. It was directly quenched in liquid nitrogen with two tweezers to obtain the section morphology, and then the temperature of the sample table in the freeze-drying oven was set at −30 °C for 3 days. XRD (Malvern PANalytical Aeris, Netherlands) was used to characterize the crystal structure of the double network hydrogels. The diffraction angle (2θ) range of the test was 10°–80°, and the scanning speed was

10°/min. The data of the diffraction peak were collected. An ultraviolet–visible spectrophotometer (Frontier Technology, China) was used to measure the transparency of the double network hydrogels. The hydrogels were placed into the instrument, and their transmittance in the visible range was measured under the wavelength of 400–800 nm.

2.4. Mechanical properties of PVA/CS/emodin hydrogel

The mechanical properties of the hydrogels were measured by an electronic universal testing machine (SUST, Zhuhai) with a 10 N load cell. Hydrogel samples were cut into dumbbell shapes (gauge length: 12 mm; total length: 35 mm, inner wide: 2 mm) to test the mechanical properties in accordance with GB/T-528 (2009). The dumbbell samples were fixed at both ends on the stretcher, and then the samples were stretched at 50 mm/min. The tensile strength was defined as the force at the breaking point divided by the cross-sectional area of the sample, and the fracture strain was defined as the displacement at the breaking point divided by the gauge length of the sample (i.e., 12 mm). The elastic modulus of the hydrogel was obtained by the slope of the tensile stress–strain in the 5 %–10 % segment. The initial 0 %–5 % segment was ignored to avoid errors caused by the shaking of the sensor in the tensile test. The toughness of the hydrogel was defined as the area of tensile stress–strain curve. Each mechanical test of hydrogel was repeated at least three times to reduce the error.

2.5. Emodin releasing of PVA/CS/emodin hydrogel

The double network hydrogels were placed in 10 mL PBS buffer and oscillated at 37 °C to simulate the human environment. The absorbance at 315 nm was measured by the UV–visible spectrophotometer at 0.5, 1, 2, 3, and 4 h separately. The initial loading amount of emodin in the hydrogel was measured to calculate the release rate of emodin. The hydrogel was shaken in ethanol for 48 h to ensure that emodin could be completely dissolved in the ethanol solution. The absorbance at 315 nm was measured, and then the release amount of emodin at different times was calculated in accordance with the standard curve as follows: release rate = (released amount/total amount) × 100 %.

2.6. In-vitro cytocompatibility

CCK-8 assay: NHDF cells were inoculated in 96-well plates, with 8×10^3 cells/well, and incubated overnight. After adherent growth, the hydrogel extract was added for 24–48 h, and then CCK-8 solution was added in accordance with the reagent instructions. After 2 h, the absorbance was measured at 450 nm by using an enzyme-labeled instrument (Thermo Fisher Scientific, USA). The larger the optical density (OD) was, the more the cell proliferation, and the stronger the cell vitality. The relative cell viability was calculated using the following formula: cell viability % = $[(A_s - A_b)/(A_c - A_b)] \times 100$ %, where A_s is the OD of the experimental well (medium + cell + drug + CCK-8), A_c is the OD of the control well (medium + cell + CCK-8), A_b is the OD of the blank well (medium + CCK-8).

Live/dead cell staining: NHDF cells were inoculated in 96-well plates, with 8×10^3 cells/well, and incubated overnight. After adherent growth, the hydrogel extract was added for continued incubation for 24–48 h, and Calcein AM/PI working solution was added proportionally. The staining effect was observed under a fluorescent inverted microscope (Carl Zeiss, USA) after incubation for 40 min under protection from light (the living cells were stained with Calcein AM and showed green fluorescence, and the dead cells was stained propyl iodide and showed red fluorescence).

2.7. Scratch wound healing assay

Scratch wound healing assay was applied to evaluate the effect of the PVA/CS/emodin hydrogel on the migration ability of NHDF cells. The cells were first inoculated in six-well plates and incubated until cell confluency reached 80 %–90 %. The cell medium was changed to serum-free medium and continued to be incubated overnight. The cells were scratched directly at the bottom of the six-well plate with the tip of a 200 μL pipette, and cell fragments were cleaned with sterile PBS. Then, different concentrations of hydrogel extracts were added, while serum-free medium was used as the control group. The scratched areas were observed under an inverted microscope at 12, 24, and 48 h separately. ImageJ software was used to calculate the area of the scratch area, and the scratch healing rate was calculated using the following formula: scratch healing rate (%) = $(1 - A_n/A_0) \times 100$ %, where A_0 and A_n are the area of the scratch area 0 and n hours after the treatment of hydrogel extract, respectively ($n = 12$, 24, or 48).

2.8. In-vitro analysis of growth factor

Quantitative real-time PCR (qRT-PCR) was used to detect the effect of the PVA/CS/emodin hydrogel dressing on the NHDF cell growth factors EGF, VEGF-A, TGF-β1, and bFGF. The NHDF cells were inoculated in six-well plates, with 1×10^6 cells per well, and incubated overnight. The hydrogel extract was added after cell confluency reached 60 %–70 % under the microscope. The total RNA from the NHDF cells was extracted for concentration determination after 48 h, and then cDNA was obtained by reverse transcription reaction. SYBR Green I chimeric dye method was used to conduct qRT-PCR. The reaction system was prepared in accordance with the kit instructions, and the qRT-RCR experiment was conducted in accordance with the reaction procedure. The C_t values of each sample were obtained after the reaction, and the relative mRNA expression was calculated by the formula $2^{-\Delta\Delta Ct}$. The primer sequence used is shown in Table S1.

2.9. In-vivo wound healing in rats

The animal research was approved by the Animal Experiment Ethics Committee of Anhui Medical University. Male SPF-grade SD rats with a body weight of 200 ± 20 g were selected for in0vivo healing experiment. The rats were fasted without water 12 h before modeling. A full-thickness skin defect model with a diameter of about 8 mm on the back was generated under anesthesia. After modeling, the rats were randomly divided into four groups: control, PVA/CS/E_0, PVA/CS/E_1, and PVA/CS/E_2. The control group was covered with sterile gauze infiltrated with normal saline, whereas the other groups were covered with sterilized hydrogel dressing, and the wound was fixed with medical elastic net bandage. All the rats in each group were photographed at 0, 3, 7, 10, and 13 days after the modeling. The wound healing degree and healing time were recorded. ImageJ was used to analyze the wound and calculate the wound healing rate. The calculation formula was as follows: wound healing rate (%) = (initial wound area − wound area on observation day)/initial wound area × 100 %.

2.10. Histopathology and α-SMA, CD31, and Ki67expression levels in wound tissues

After the modeling was successfully conducted for 3, 7, and 13 days, the rats in each group were sacrificed, sampled, and fixed in 4 % neutral formalin solution immediately. After paraffin embedding was performed, the tissues were sliced at a thickness of 4 μm. HE staining and Masson staining were then performed using tissue sections in accordance with the kit instructions. HE staining was used to evaluate the pathological changes and inflammation reaction of the wound, and Masson staining was used to evaluate the collagen deposition of the wound. Then, in accordance with the kit instructions, the wound tissue was stained by immunofluorescence with anti-α-SMA antibody, and immunohistochemical analysis was performed with anti-CD31 and Ki67 antibodies separately.

2.11. Statistical analysis

Data were expressed as the mean ± standard deviation when they conformed to a normal distribution. Meanwhile, they were analyzed by ANOVA followed by Bonferroni test for multiple comparisons. The difference was statistically significant at $p < 0.05$.

3. Results and discussion

3.1. Preparation of PVA/CS/emodin hydrogel

As shown in Fig. 1A, the polymer film obtained by the dehydration of the PVA and CS mixed solution in a drying oven was transparent, hard, and brittle. In the process of solution dehydration, the precipitation of water molecules led to the proximity of PVA molecular chains, and the aggregation of hydroxyl groups on the molecular side chains led to the crystallization of PVA, finally forming a highly crystalline polymer film. When soaking the solution, the water molecules greatly destroyed the crystallization of PVA and entered between the molecular chains of PVA. Then, the water molecules formed hydrogen bonds with hydrophilic polymer chains, and the polymer film turned into a hydrogel in the process. After the water molecules entered the gel network, the crystallinity was damaged, and the mesh of the hydrogel was expanded. The relatively weak crystalline structure of the PVA molecules formed a network of the hydrogel. Meanwhile, emodin entered into the network with water molecules, and two carbonyl groups of emodin molecules reacted with amino groups on the CS molecules as Schiff base addition reaction to form a dynamic covalent crosslinked network. The dynamic covalent bond between emodin and CS formed a second network inside the hydrogel, successfully constructing the structure of the double network hydrogel.

The establishment of a double network structure played a key role in the mechanical properties of gels, and the successful loading of emodin into the hydrogel endowed it with biological properties. As shown in Fig. 1B, emodin could promote cell proliferation, cell migration, and secretion of growth factors at the cellular level and accelerate wound healing at the animal experimental level by facilitating the regeneration of blood vessels and collagen. Meanwhile, the successful loading of emodin into the hydrogels endowed them with biological properties.

3.2. Characterization of double network hydrogel

The structure of the hydrogel was characterized, and the mechanical properties was measured to verify the double network structure of the

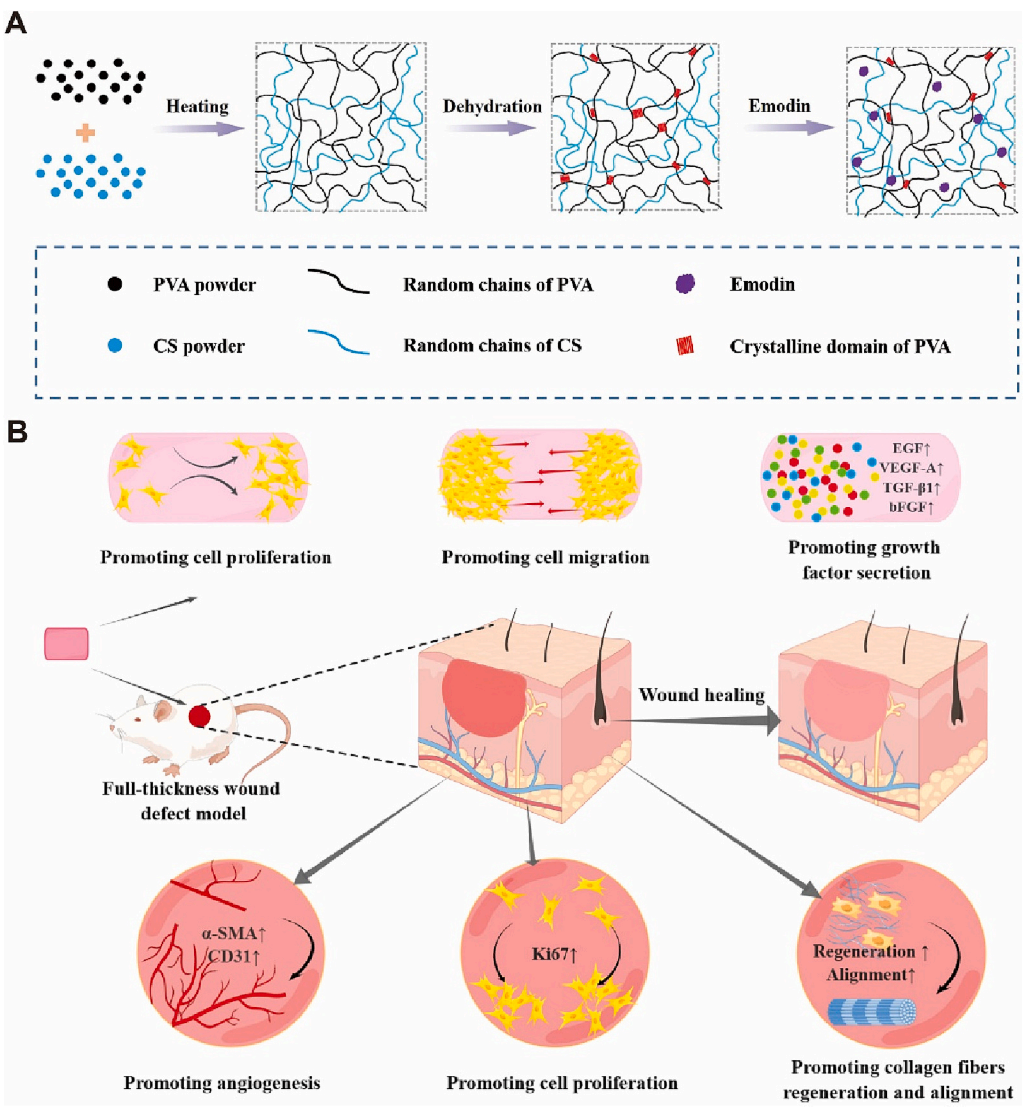

Fig. 1. Schematic of the fabrication and properties of PVA/CS/emodin hydrogels. (A) Diagram of hydrogel synthesis. (B) Diagram of the effects on hydrogels in vivo and in vitro.

hydrogel. As shown in Fig. 2A, the SEM image of the polymer film without soaking solution showed obvious molecular chain aggregates but no obvious network structure. Fig. 2B illustrates the apparent mesh structure of the hydrogel after soaking in emodin solution. By contrast, the process of soaking solution significantly promoted the formation of hydrogel network, whereas the aggregation of molecular chains decreased it correspondingly. The SEM photos of PVA/CS, PVA/CS/E_1, PVA/CS/E_2 and PVA/CS/E_3 are shown in **Fig. S1**. With the addition of a small amount of emodin, PVA/CS/E_1 demonstrated a semi-mesh structure that was not very obvious, while PVA/CS/E_2 showed a distinctly visible network structure. However, with the further increase of emodin concentration, the mesh structure of PVA/CS/E_3 was slightly diminished with uneven distribution. This could be attributed to the fact that the higher emodin concentration improved the cross-linking degree of the network. XRD tests were further conducted on the polymer film and hydrogel. As shown in Fig. 2C, the polymer film showed a very obvious wide diffraction peak when 2θ was 20° and a small diffraction peak when 2θ was 40°, whereas the diffraction peak of hydrogel decreased greatly at 20°, and a narrow and sharp diffraction peak appeared at 30°. This finding indicated that the entry of water molecules destroyed the crystallinity of the polymer and formed a weak crystalline network structure during the formation of hydrogel. Fig. 2D shows the transparency of hydrogels containing emodin of different concentrations in the visible range of 400–800 nm, indicating that the transmittance of hydrogels decreased with the increase in emodin. The transparency of the PVA/CS/E_0 hydrogels was >90 % above 680 nm, whereas that of the PVA/CS/E_1 group showed a slight decrease, but the transmittance was also >87 %. For the PVA/CS/E_2 and PVA/CS/E_3 groups, the transmittance of emodin outside the red range (400–630 nm) decreased with the increase in emodin concentration, whereas the transmittance in the red range remained around 90 %. Considering that the emodin solution has a clear red color, this finding also proved that emodin is involved in the formation of the hydrogel network.

3.3. Mechanical properties of double network hydrogel

As the measurement of mechanical properties demonstrated that emodin is involved in the formation of hydrogel network, the mechanical properties of hydrogel could be greatly adjusted by changing the emodin concentration. Fig. 3A shows the typical tensile stress–strain curves of PVA/CS/E_1, PVA/CS/E_2, and PVA/CS/E_3 hydrogels. The tensile strength and fracture strain of these three hydrogels increased with the increase in emodin concentration. The tensile strength values of the PVA/CS/E_1, PVA/CS/E_2, and PVA/CS/E_3 hydrogels were 325, 635, and 1070 kPa, respectively, and their fracture strains were 76 %, 110 %, and 154 %, respectively. By contrast, the tensile strength values of the PVA/CS/E_2 and PVA/CS/E_3 groups were 1.9 and 3.3 times higher than that of the PVA/CS/E_1 group, respectively, and the fracture strain values were 1.4 and 2.0 times higher, respectively. The elastic modulus of the hydrogel also increased with the addition of emodin. As shown in Fig. 3B, the elastic modulus values of the PVA/CS/E_1, PVA/CS/E_2, and PVA/CS/E_3 groups were 396, 443, and 638 kPa, respectively. This finding showed that the modulus of the hydrogels was appropriate, and when applied in wound dressing, they could fit the wound well, thus avoiding the contact between the wound and the outside air and reducing the risk of bacterial infection. Fig. 3C shows the toughness of hydrogels by calculating the area of the stress–strain curve. The toughness values of the PVA/CS/E_1, PVA/CS/E_2, and PVA/CS/E_3 groups were 132, 325, and 803 kJ/m^3, respectively. The excellent toughness ensured the integrity of hydrogel as a wound dressing.

Therefore, emodin was a component of the hydrogel network. However, it could also be released in subsequent treatment because it was excessive during soaking. Fig. 3D shows the drug release experiment of the hydrogel to verify its emodin release ability. The emodin releases of PVA/CS/E_1, PVA/CS/E_2 and PVA/CS/E_3 hydrogels were fast in the initial stage, the drug release rates reached 12.4 %, 49.3 % and 67.6 % in half an hour. With the increase in time, the drug release rate slowed down, and the drug release rates of PVA/CS/E_1, PVA/CS/E_2 and PVA/CS/E_3 at 4 h were 20.3 %, 77.2 % and 84.4 %, respectively. By 4 h, the emodin release process was almost complete. The release rate of emodin increased with the increase of emodin concentration for PVA/CS/E_1, PVA/CS/E_2 and PVA/CS/E_3, which can be attributed to the fact that the content of free emodin inside the network rose with the increase of emodin concentration due to the constant content of chitosan for these three hydrogels. Moreover, in order to ensure the solubility of emodin in PBS solution, PVA/CS/E_1, PVA/CS/E_2 and PVA/CS/E_3 solutions after

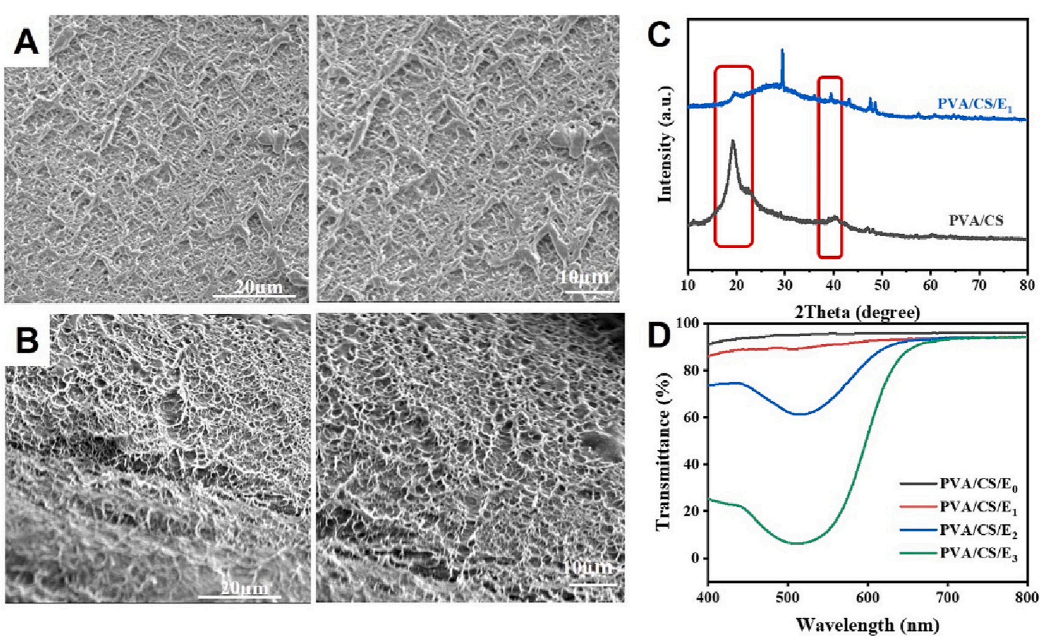

Fig. 2. Structure and characterization of hydrogels. (A) SEM image of the polymer film without soaking solution. (B) SEM image of the hydrogel after soaking in emodin solution. (C) Intensities of hydrogels in PVA/CS and PVA/CS/E_1 groups. (D) Transmittance of hydrogels in PVA/CS/E_0, PVA/CS/E_1 PVA/CS/E_2, and PVA/CS/E_3 groups.

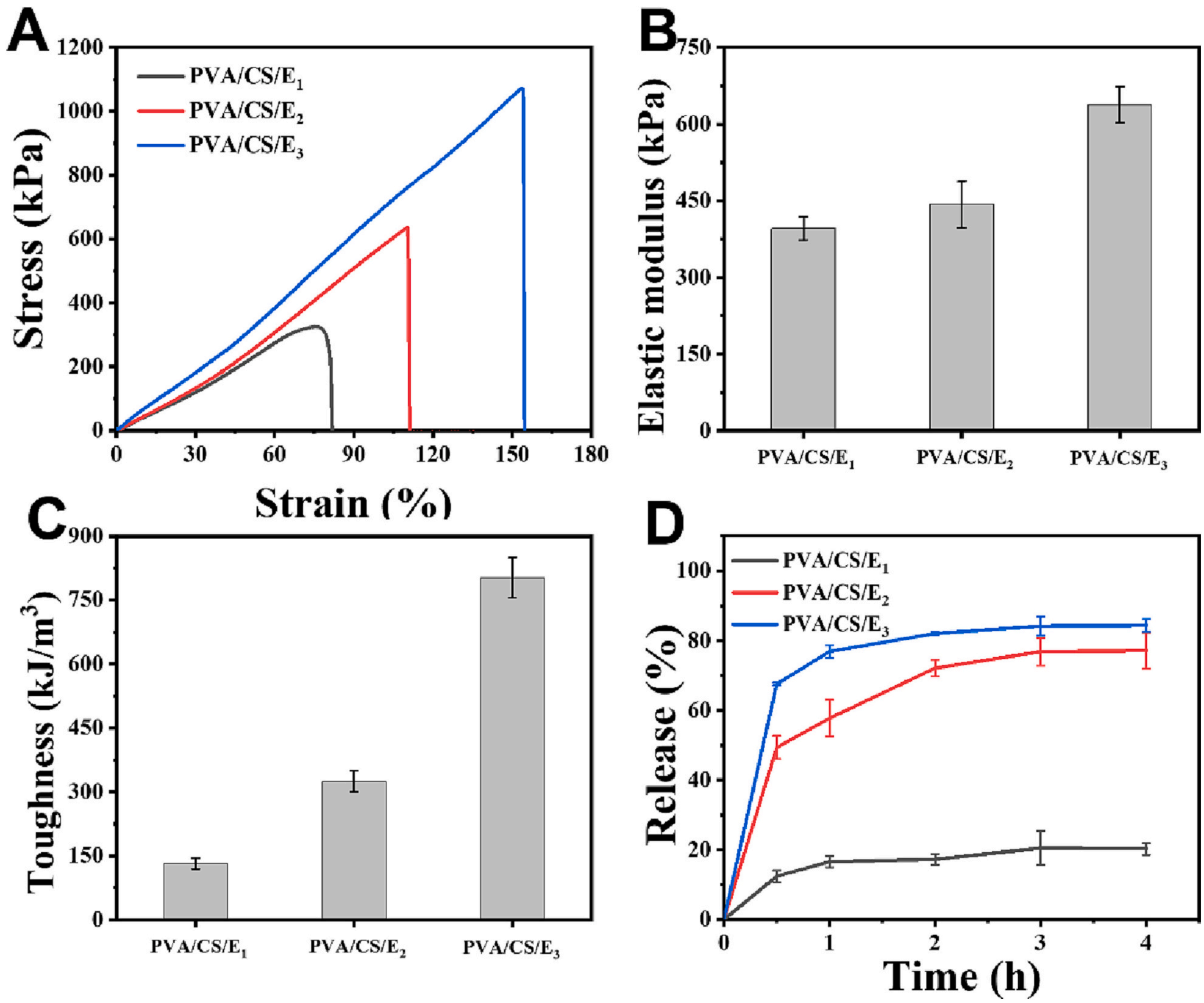

Fig. 3. Mechanical properties and emodin release of hydrogels. (A) Typical tensile stress–strain curves of PVA/CS/E_1, PVA/CS/E_2 and PVA/CS/E_3 hydrogels. (B) Elastic modulus of hydrogels in PVA/CS/E_1, PVA/CS/E_2, and PVA/CS/E_3 groups. (C) Toughness of hydrogels in PVA/CS/E_1, PVA/CS/E_2, and PVA/CS/E_3 groups. (D) Drug release ability of emodin in PVA/CS/E_2 group.

release were left to rest for three days. As shown in Fig. S2, we have not found precipitation of emodin, showing a homogeneous solution.

3.4. Cytocompatibility and migration test of hydrogel

As one of the most important indicators to evaluate the safety of biological materials, biocompatibility is often verified prior to clinical application of wound dressings. In this study, the biocompatibility of the double network hydrogel loaded with emodin was evaluated by CCK-8 assay and live/dead cell staining. Fig. 4A shows the cell viability of NHDFs in different groups at 24 and 48 h. All the groups showed good biocompatibility in 24 h, and the cell viability of the first four groups was above 85 %. In 48 h, the cell viability of the first four groups reached >95 %. In particular, the PVA/CS/$E_{0.1}$, PVA/CS/E_1, and PVA/CS/E_2 groups reached >115 %. The cell viability of groups loaded with lower concentrations of emodin was higher than that of the group without emodin. The PVA/CS/$E_{0.1}$, PVA/CS/E_1, and PVA/CS/E_2 groups all had a certain proliferative effect on cell growth, and this effect increased with the addition of emodin. However, when the emodin concentration was too high, the PVA/CS/E_3 hydrogels also had an obvious inhibitory effect on cell growth, and cell viability decreased significantly. Fig. 4C shows the live cell staining of NHDFs in different groups at 12, 24, and 48 h. The NHDF cells in all groups grew normally and were in a good state of polygonal or spindle shape at 12 h, except those in the PVA/CS/E_3 group at 24 h, which showed a remarkable decrease in the number of living cells. The number of living cells in the PVA/CS/E_3 group decreased significantly at 48 h, whereas that in the other groups increased significantly compared with that in the control group. Therefore, the CCK-8 and live/dead cell staining experiments confirmed that the hydrogel dressing of each group had good biocompatibility except that of the PVA/CS/E_3 group. Meanwhile, the hydrogel dressing of the PVA/CS/$E_{0.1}$, PVA/CS/E_1, and PVA/CS/E_2 groups had certain ability to promote cell proliferation. The dead cell staining of NHDFs in different groups at 12, 24, and 48 h is shown in Fig. S3.

Cell migration is one of the necessary processes of body repair, regeneration, and wound healing. In this study, the process of cell migration was simulated in vivo. Fig. 4B shows the scratch healing rate of NHDFs in different groups, and Fig. 4D reflects the cell migration of the hydrogel in different groups. The cell migration of the PVA/CS/E_0, PVA/CS/$E_{0.1}$, PVA/CS/E_1, and PVA/CS/E_2 groups slightly increased, and the cell scratch healing rate was higher than that in the control group at 24 h. The cell migration of the PVA/CS/$E_{0.1}$, PVA/CS/E_1, and PVA/CS/E_2 groups gradually increased with the addition of emodin at 48 h, and the cell scratch healing rate also increased gradually, whereas no significant change was found in the cell migration of the PVA/CS/E_3 group. Therefore, the hydrogel dressings of the PVA/CS/$E_{0.1}$, PVA/CS/

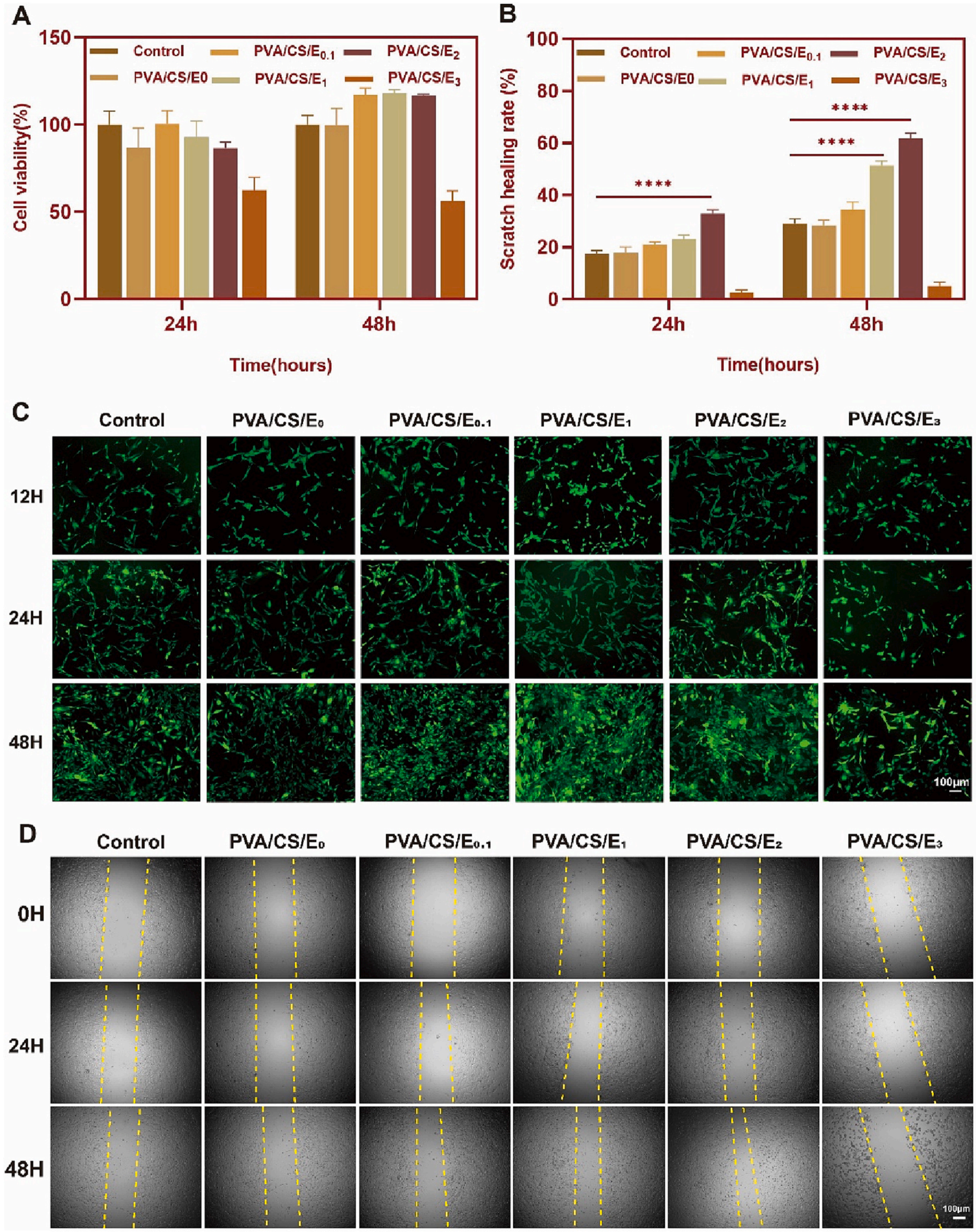

Fig. 4. Cytocompatibility and migration test of hydrogels. (A) Cell viability of NHDFs in different groups at 24 and 48 h. (B) Scratch healing rate of NHDFs in different groups at 24 and 48 h (****$P < 0.0001$, PVA/CS/E_2 vs. control group at 24 h; ****$P < 0.0001$, PVA/CS/E_1 and PVA/CS/E_2 vs. control group at 48 h). (C) Live cell staining of NHDFs in different groups at 12, 24, and 48 h. (D) Cell migration of NHDFs in different groups at 0, 24, and 48 h.

E_1, and PVA/CS/E_2 groups demonstrated good ability to promote cell migration.

3.5. Effect of PVA/CS/emodin hydrogel on mRNA expression of growth factor

Emodin could promote wound healing, so the double network hydrogel dressing loaded with emodin was speculated to have the ability to improve the release of cytokines related to wound healing. Fig. 5 shows the mRNA relative expression levels of growth factors (EGF, VEGF-A, TGF-β1, and bFGF) related to wound healing. In this experiment, the PVA/CS/E_2 group, which had better biocompatibility than the experimental group, was selected. The results confirmed that the mRNA relative expressions levels of EGF (Fig. 5A), VEGF-A (Fig. 5B), TGF-β1 (Fig. 5C), and bFGF (Fig. 5D) in the PVA/CS/E_2 group were significantly higher than those in the control group, and the differences were statistically significant. The mRNA expression of TGF-β1 in the PVA/CS/E_0 group was significantly higher than that in the control group, and the difference was statistically significant, whereas the mRNA expression levels of the other three cytokines were not significantly different from those in the control group. Therefore, the PVA/CS/emodin hydrogel dressing could facilitate the expression of growth factors related to wound healing and angiogenesis.

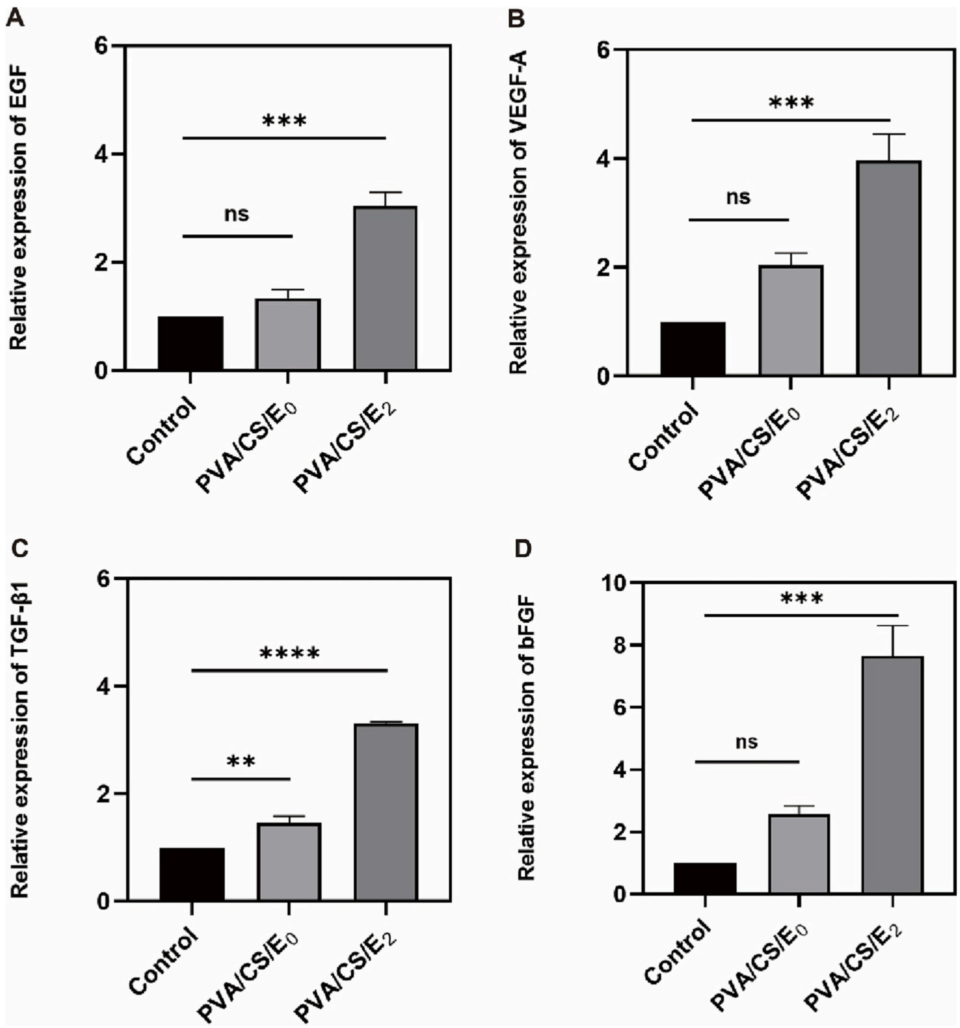

Fig. 5. mRNA relative expression levels of growth factors in control, PVA/CS/E_0, and PVA/CS/E_2 groups at 48 h. (A) Relative expression of EGF (***$P < 0.001$, PVA/CS/E_2 vs. control group). (B) Relative expression of VEGF-A (***$P < 0.001$, PVA/CS/E_2 vs. control group). (C) Relative expression of TGF-β1 (****$P < 0.0001$, PVA/CS/E_2 vs. control group; **$P < 0.01$, PVA/CS/E_0 vs. control group). (D) Relative expression of bFGF (***$P < 0.001$, PVA/CS/E_2 vs. control group).

3.6. In-vivo wound healing evaluation of PVA/CS/emodin hydrogel

The above experiments confirmed that the PVA/CS/emodin hydrogel had good biocompatibility, which could meet the application standards as medical biological dressing. Thus, experiments were further carried out on animal wounds, and the PVA/CS/E_1 and PVA/CS/E_2 groups, which had better biocompatibility than the experimental group, were selected. As shown in Fig. 6A, acute wounds were observed on the rats immediately after modeling and confirmed to be full-thickness skin defects. The wounds in the four groups were gradually reduced with the passage of time. At the same time, the granulation tissue of the PVA/CS/E_1 and PVA/CS/E_2 groups grew faster than that of the control and PVA/CS/E_0 groups, and the wound reduction was more obvious. At 13 days after modeling, the wounds of the PVA/CS/E_1 and PVA/CS/E_2 groups basically healed, whereas those of the control and PVA/CS/E_0.

groups still had some residual wounds. Fig. 6B shows the schematic of the wound residual area at different time points in the four groups of rats. The acute wound healing rate of the rats was also recorded and statistically analyzed, as shown in Fig. 6C. At 3 days after modeling, the wound healing rate of the PVA/CS/E_1 and PVA/CS/E_2 groups significantly increased compared with that of the control group. The wound healing rate of the PVA/CS/E_2 group reached 37.1 %, whereas that the control group was 15.1 %. At 7 days after modeling, the wound healing rate of the PVA/CS/E_1 and PVA/CS/E_2 group reached 68.1 % and 77.6 %, respectively, compared with 47.0 % in the control group. The wound healing rate of the PVA/CS/E_1 and PVA/CS/E_2 groups increased significantly at 10 days after modeling. The wound healing rate of the PVA/CS/E_2 group reached 96.1 %, and the increase was more obvious and close to healing, whereas the wound healing rate of the PVA/CS/E_0 group was not significantly higher than that of the control group. At 13 days after modeling, the wound healing rate of the PVA/CS/E_1 and PVA/CS/E_2 groups reached 99.2 % and 100 %, respectively, and the wounds were basically completely healed. In addition, the healing time from acute wound modeling to complete epithelialization in the rats was recorded and counted. Fig. 6D shows the change of wound healing time in the rats. The mean healing times of the PVA/CS/E_1 and PVA/CS/E_2 groups were 13.4 and 12.4 days compared with 15.4 days in the control group. In conclusion, the wound healing rates of the PVA/CS/E_1 and PVA/CS/E_2 groups significantly increased compared with that of the control group, and the healing time significantly shortened.

In addition, the histomorphology of the wound tissue during the healing process of the full-thickness defect model rats was observed. Fig. 7A shows the pathological changes observed by HE staining. At 3, 7, and 13 days after modeling, the wound tissue of the PVA/CS/E_1 and PVA/CS/E_2 groups showed relatively fewer inflammatory cells than that of the control group. The new granulation tissue grew faster, and the interstitial edema was also relieved faster, which promoted the formation of neovascularization and accelerated the formation of new

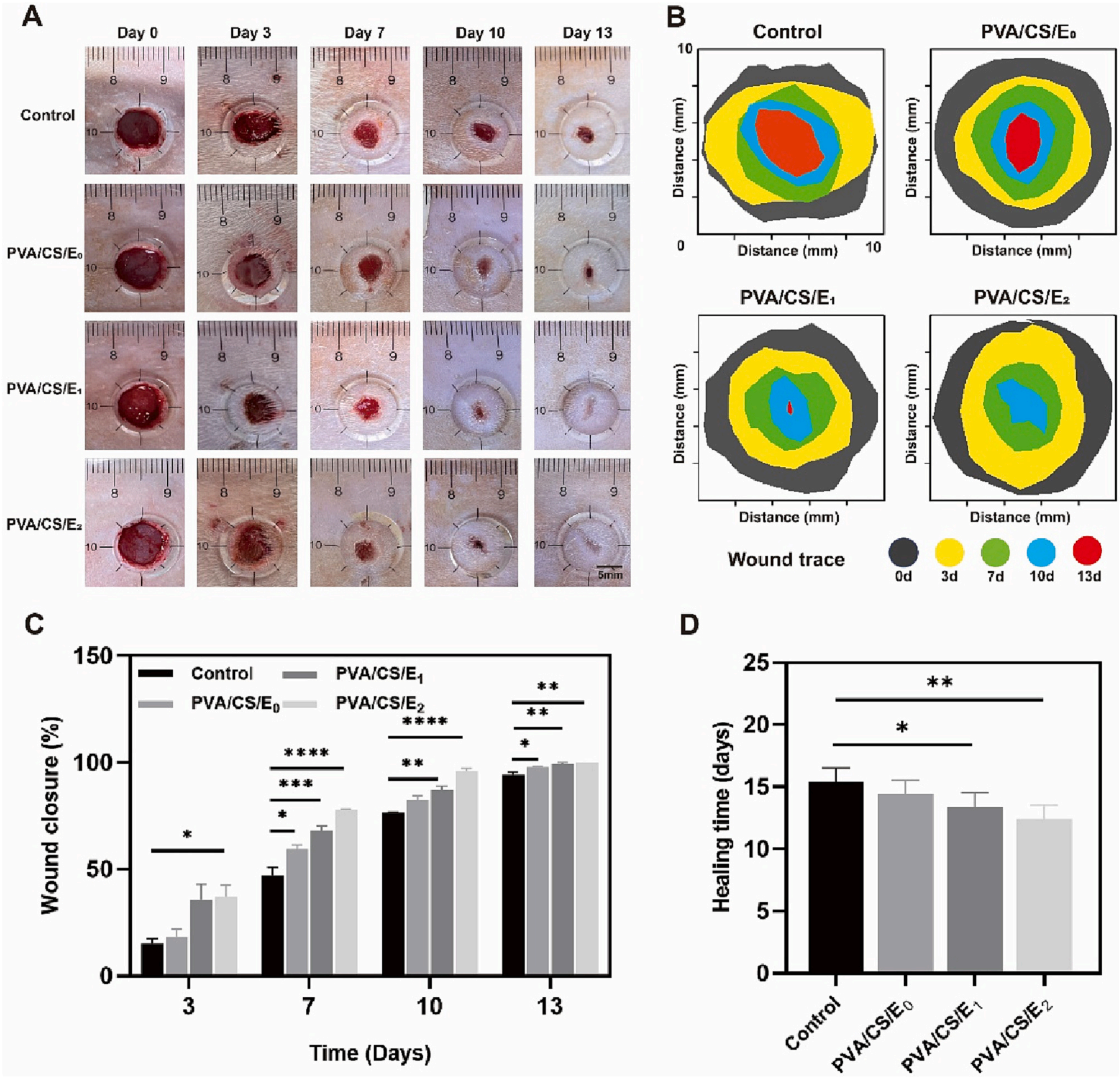

Fig. 6. In-vivo wound healing evaluation of PVA/CS/emodin hydrogels in control, PVA/CS/E_0, PVA/CS/E_1, and PVA/CS/E_2 groups. (A) Wound healing of rats in different groups at 0, 3, 7, 10, and 13 d. (B) Schematic of residual wound area at 0, 3, 7, 10, and 13 d. (C) Wound closure rate of rats in different groups at 3, 7, 10, and 13 d after modeling (at 3 d, $^*P < 0.05$, PVA/CS/E_2 vs. control group. At 7 d, $^*P < 0.05$, PVA/CS/E_0 vs. control group, $^{***}P < 0.001$, PVA/CS/E_1 vs. control group, $^{****}P < 0.0001$, and PVA/CS/E_2 vs. control group. At 10 d, $^{**}P < 0.01$, PVA/CS/E_1 vs. control group, $^{****}P < 0.0001$, PVA/CS/E_2 vs. control group. At 13 d, $^*P < 0.05$, PVA/CS/E_0 vs. control group, $^{**}P < 0.01$, PVA/CS/E_1 vs. control group, and $^{**}P < 0.01$, PVA/CS/E_2 vs. control group). (D) Wound healing time of rats in different groups ($^*P < 0.05$, PVA/CS/E_1 vs. control group. $^{**}P < 0.01$, PVA/CS/E_2 vs. control group).

epithelium. Meanwhile, Masson staining was also used to observe the changes in collagen fibers, as shown in Fig. 7B. Collagen deposition was found to be increased in the wound tissues of the PVA/CS/E_1 and PVA/CS/E_2 groups compared with the control group after modeling. This phenomenon promoted the maturation and proliferation of collagen fibers, improved the arrangement of collagen, and thus accelerated the rate of wound epithelization. Therefore, the PVA/CS/emodin double network hydrogel dressing demonstrated an obvious ability to accelerate wound healing.

3.7. In-vivo α-SMA, CD31, and Ki67 expression levels in wound tissue of PVA/CS/emodin hydrogel

The above experiments confirmed that the PVA/CS/emodin double network hydrogel dressing could accelerate wound healing. Then, the expression levels of α-SMA, CD31, and Ki67 in wound tissue were further detected by immunofluorescence and immunohistochemistry to study the mechanism of promoting healing. α-SMA is involved in angiogenesis and is commonly used to mark mature blood vessels, playing an important role in the process of wound healing. As shown in Fig. 8A, the α-SMA on the acute wounds of rats was labeled by immunofluorescence at 3, 7, and 13 days after modeling and observed under the fluorescence microscope. The expression levels of α-SMA in the

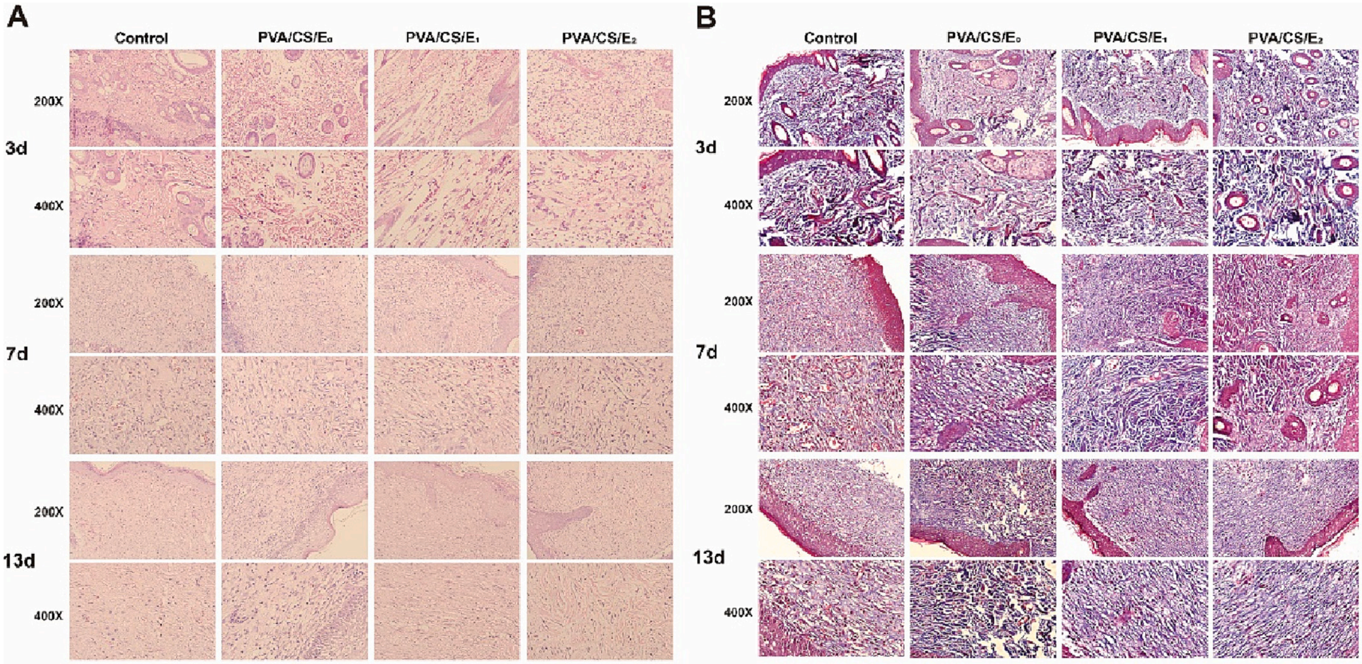

Fig. 7. Histomorphological observations of PVA/CS/emodin hydrogels in control, PVA/CS/E_0, PVA/CS/E_1, and PVA/CS/E_2 groups. (A) HE staining of rats in different groups at 3, 7, and 13 d after modeling. (B) Masson staining of rats in in different groups at 3, 7, and 13 d after modeling.

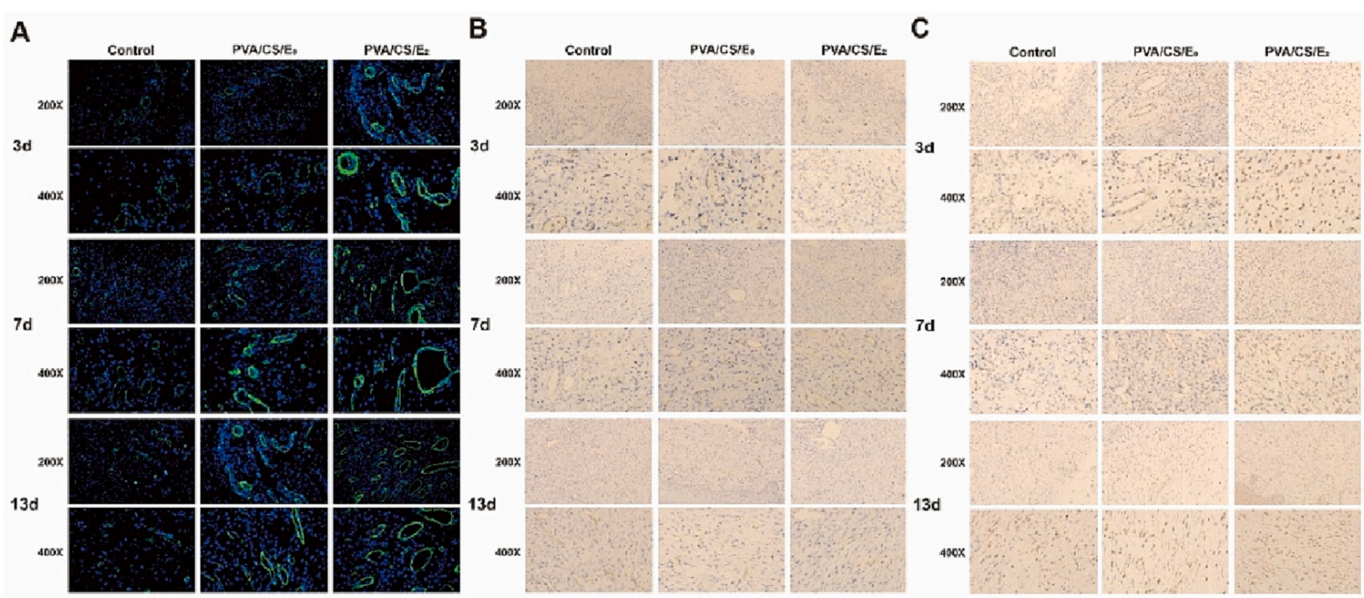

Fig. 8. In-vivo α-SMA, CD31 and Ki67 expression in wound tissue of PVA/CS/emodin hydrogel in control, PVA/CS/E_0, and PVA/CS/E_2 groups. (A) IF expression of α-SMA in different groups at 3, 7, and 13 d after modeling. (B) IHC expression of CD31 in different groups at 3, 7, and 13 d after modeling. (C) IHC expression of Ki67 in different groups at 3, 7, and 13 d after modeling.

control, PVA/CS/E_0, and PVA/CS/E_2 groups increased gradually over time. At 3 days after modeling, no significant difference was found between the groups for α-SMA expression. At 7 days and 13 days after modeling, the fluorescence expression of α-SMA in the PVA/CS/E_0 group was slightly higher than that in the control group at the same time, whereas that in the PVA/CS/E_2 group was significantly higher than that in control group.

CD31, which is a platelet-endothelial cell adhesion molecule, is involved in angiogenesis and plays an important role in wound healing. It is a marker of vascular endothelium and is commonly used to mark new blood vessels. As shown in Fig. 8B, the CD31 on the acute wounds of rats was labeled and observed by immunohistochemistry at 3, 7, and 13 days after modeling. The expression levels of CD31 in the control, PVA/CS/E_0, and PVA/CS/E_2 groups all increased gradually over time. At 3 days after modeling, the CD31 expression in the three groups decreased, and no significant difference was observed. At 7 and 13 days after modeling, the expression of CD31 in the PVA/CS/E_0 group was slightly higher than that in the control group at the same time, whereas that in the PVA/CS/E_2 group was significantly higher than that in the control group at the same time.

Ki67 is an antigen associated with proliferating cells, whose function is related to mitosis and indispensable in cell proliferation. It is commonly used to label cells in the proliferative cycle. As shown in Fig. 8C, the Ki67 on the acute wounds of rats was labeled by immunohistochemistry at 3, 7, and 13 days after modeling and observed under the microscope. The Ki67 expression levels in the control, PVA/CS/E_0, and PVA/CS/E_2 groups also increased gradually over time. At 3 days after modeling, the Ki67 expression in the three groups decreased, without significant difference. At 7 and 13 days after modeling, the expression of Ki67 in the PVA/CS/E_2 group was significantly higher than that in the control group at the same time.

To sum up, the results of immunofluorescence and immunohistochemistry showed that the expression levels of α-SMA, CD31, and Ki67 in the PVA/CS/E_2 group containing emodin were significantly higher than those in the control and PVA/CS/E_0 groups without emodin. Moreover, the PVA/CS/emodin double network hydrogel dressing demonstrated good ability to promote wound healing.

4. Conclusions

In the process of wound healing, dressing could resist external infection by covering the wound, provide a temporary barrier for wounds and guide skin reconstruction, which plays an important role in wound healing. Aiming at the defects of traditional wound healing dressing, this study designs a double network hydrogel dressing containing emodin to accelerate wound healing process based on the moisturizing property, mechanical property, biocompatibility and healing promotion performance of the dressing. Specifically, a double network hydrogel based on the CS–emodin network formed by Schiff base reaction and microcrystalline PVA network was constructed. The double network structure endowed the hydrogel with excellent mechanical properties and ensured its integrity as a wound dressing. The tensile strength of the hydrogel increased from 325 kPa to 1070 kPa. The toughness also increased by six times with the increase in emodin concentration. Moreover, emodin not only acted as a component of the network to improve the mechanical properties of the hydrogel, but also could release drugs to promote the wound. At the cellular level, the hydrogel dressing could promote cell proliferation and migration. It could also enhance the expressions levels of growth factors (EGF, VEGF-A, TGF-β1, and bFGF) related to wound healing. In the in-vivo wound healing experiment, the hydrogel dressing effectively accelerated wound healing. The results of the expression levels of α-SMA, CD31, and Ki67 in the wound tissue by immunofluorescence and immunohistochemistry also demonstrated that the hydrogel dressing facilitated the regeneration of blood vessels and collagen.

CRediT authorship contribution statement

JW: Conceptualization, Experiments, Data Curation, Software, Investigation, Formal Analysis, Writing - Original Draft.

YL: Conceptualization, Experiments, Data Curation, Software, Investigation, Formal Analysis.

XW: Experiments, Data Curation.

HL: Resources, Supervision, Writing - Review & Editing.

XC: Conceptualization, Funding Acquisition, Resources, Supervision, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2023.125610.

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